

MT 205: Functional Properties of Materials

Course Code	MT 205
Course Name	Functional Properties of Materials
L-T-P-C	3-2-0-4
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	EN509: Functional Materials for Energy Engineering
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** Overview of functional materials, including inorganic, organic, and metal-organic frameworks, as well as hybrid organic-inorganic perovskites. Discussion on the use of material functionalities in device fabrication. Exploration of functionalities driven by electronic, spin, and ionic properties, along with the integration of these effects for designing novel materials. [4 Hours]
- **Functionality Driven by Electronic Properties:** Examination of amorphous and crystalline solids, electronic states, band formation, and band dispersions in crystalline materials. Coverage of metals, semiconductors, and insulators, including direct and indirect bandgap semiconductors, impurity bands in doped semiconductors, charge carrier transport, and optical properties, with a focus on optoelectronic materials. [12 Hours]
- **Functionality Driven by Spin Properties:** Exploration of magnetic moments in atoms, spin and orbital contributions in solids, magnetization, and various types of magnetic materials and their applications. Detailed discussion on exchange interactions, magnetic transition temperatures (mean-field theory), domain wall formation, and materials like Magnetoresistance, CMR/GMR. [10 Hours]
- **Functionality Driven by Ionic Properties:** Investigation of covalent, ionic, and metallic solids, dipole formation, and polarization. Overview of paraelectric, ferroelectric, antiferroelectric, piezoelectric, and pyroelectric materials, including domain wall formation in ferroelectrics and multiferroic materials. [6 Hours]
- **Functional Devices Driven by Functionality:** Study of functional devices such as light-emitting diodes, solar cells, quantum devices, FETs, Memristors, MOS Capacitors, photodetectors, thermoelectric, and photovoltaic devices, showcasing the practical application of functional materials. [10 Hours]

Laboratory Modules

1. Crystal structures (Utilizing 3D models to understand crystal structures and understand lattice, packing fraction, mass, density, concept of Avogadro Number, etc.).
2. Optical properties characterization using absorption/photoluminescence spectroscopy.
3. Evaluate the electrical conductivity of materials through experiments such as four-point probe measurements.
4. Magnetic/Piezoelectric materials; measurement of M-H and P-E hysteresis loops.
5. Thermal Analysis: Perform experiments to measure thermal properties like thermal conductivity, TGA-DSC, specific heat, and thermal expansion of different materials.
6. Superconductivity and its phenomena. Demonstrating the Meissner effect and examining persistent currents.
7. Design and development of power packs with supercapacitors.
8. Characterizations of optoelectronic devices such as solar cells, photodetectors, and phototransistors. Analyse the effects of varying light intensity on power output of solar cells, current-voltage characteristics/ detectivity & sensitivity of the photodetector.

Text Books

1. W. Ashcroft and N.D. Mermin, **Solid State Physics**, Harcourt College Publishers, 1976.
2. Marius Grundmann, **The Physics of Semiconductors: An Introduction Including Devices and Nanophysics**, Springer, 2010.
3. W.F. Smith, **Principles of Materials Science and Engineering**, McGraw-Hill.
4. R.A. Flinn and P.K. Trojan, **Engineering Materials and Their Applications**, Haughton.
5. Neil W. Ashcroft and N. David Mermin, **Solid State Physics**, Saunders College Publishing, 1976

References

1. R. M. Martin, **Electronic Structure: Basic Theory and Practical Methods**, Cambridge University Press, 2004.
2. K.F. Wang, J. – M. Liu, and Z. F. Ren, **Multiferroicity: The coupling between magnetic and polarization orders**, Advances in Physics 58, 321, 2009.
3. S. O. Kasap, **Principles of Electronic Materials and Devices**.